

CVIS Hands-on Training on Development of Machine Learning and Deep Learning Models Summer 2025

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Inspiring Excellence



Inspiring Innovation

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Deep Learning for Video Processing Part-1



Basics

Brightness:

overall lightness or darkness of an image. It is a global adjustment that affects all pixels equally

Contrast:

Difference in intensity between the light and dark areas of an image

Luminance

refers to the perceived brightness of a color, which is the weighted sum of the RGB channels, representing human vision sensitivity.

Textures: Patterns that represent variations in pixel intensities

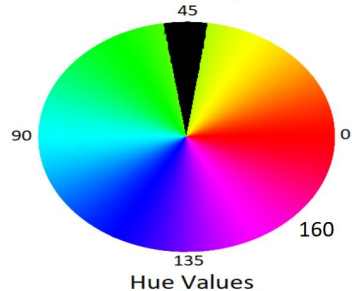


Hue represents the **color type** itself. It defines the **basic color** such as **red**, **blue**, **yellow**, etc. It is essentially the angle on the color wheel 0 to 360 degree

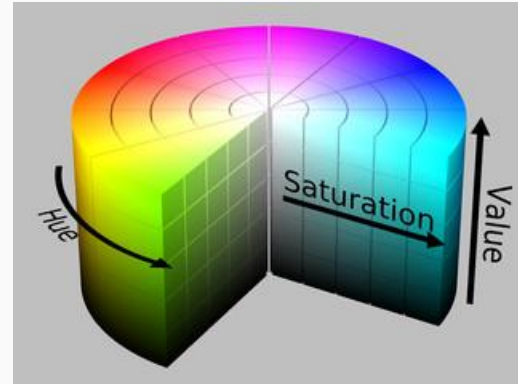
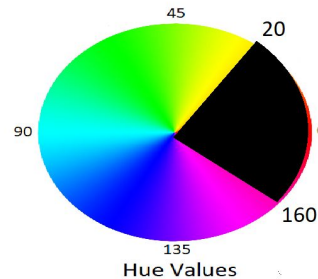
Saturation: Saturation refers to the **intensity** or **vividness** of the color. It describes how **pure** or **washed out** a color is in percent

Value: Similar to brightness 0-100%

```
image_lower_hsv = np.array([40, 200, 220])  
image_upper_hsv = np.array([50, 255, 255])
```



```
image_lower_hsv = np.array([?, 200, 220])  
image_upper_hsv = np.array([?, 255, 255])
```





FPS is the number of frames displayed per second. The higher the FPS, the smoother the video.

Common frame rates include 24, 30, and 60 FPS.

In Python, you can retrieve the FPS of a video with `cv2.VideoCapture().get(cv2.CAP_PROP_FPS)`.

Resolution

dimensions of each frame, usually given as width × height

`cap.get(cv2.CAP_PROP_FRAME_WIDTH)` and `cap.get(cv2.CAP_PROP_FRAME_HEIGHT)`

VideoCapture

`cv2.VideoCapture()` - read video or video streams/ webcam retrieve frames, process, display, save

VideoWriter

`cv2.VideoWriter()` - write processed video frames into new video file need to specify output file, codec, frame rate, frame size



Grayscale each pixel represents intensity (brightness) with no color.

`cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)`

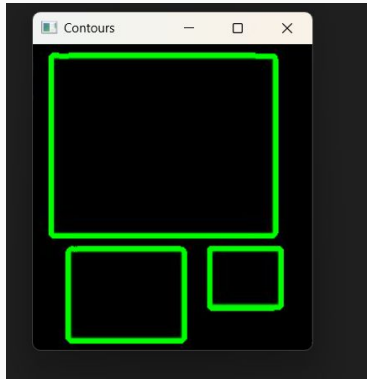
YUV color spaces

YUV separates the brightness (luminance, Y) from the color information (chrominance, U and V).

Contours

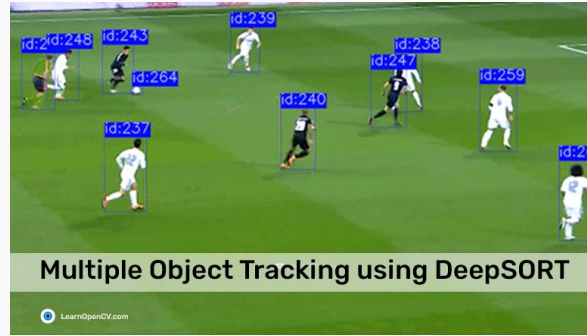
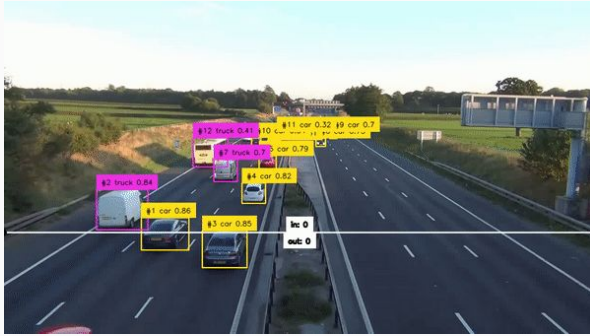
- boundaries of objects within a video frame.
- In video processing, contour detection is often used for object tracking or shape analysis.

`cv2.findContours()`.



Object Tracking

detecting and following the movement of specific objects through the frames of a video.



Algorithms:

KCF, MIL, TLD



KCF

Kernelized Correlation Filter (KCF)

Improvement of traditional Correlation Filter (CF)

Filter Learning: You learn a filter from a training image (the object).

Filtering: You apply the filter to a new frame and find the location that maximizes the correlation with the template (the object in the previous frame).

Return the peak where the object is located

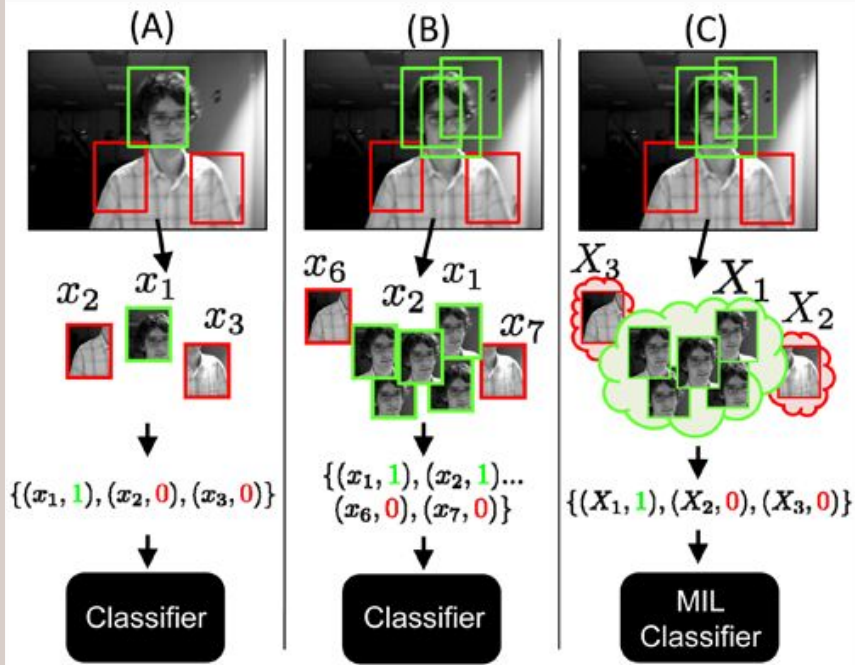
CF limitations -> learn in linear space

Steps:

1. Extract feature-> HOG Histogram of gradients
2. Solving the filter using FFT in frequency domain
3. Apply, Update and optimize
4. Optimize

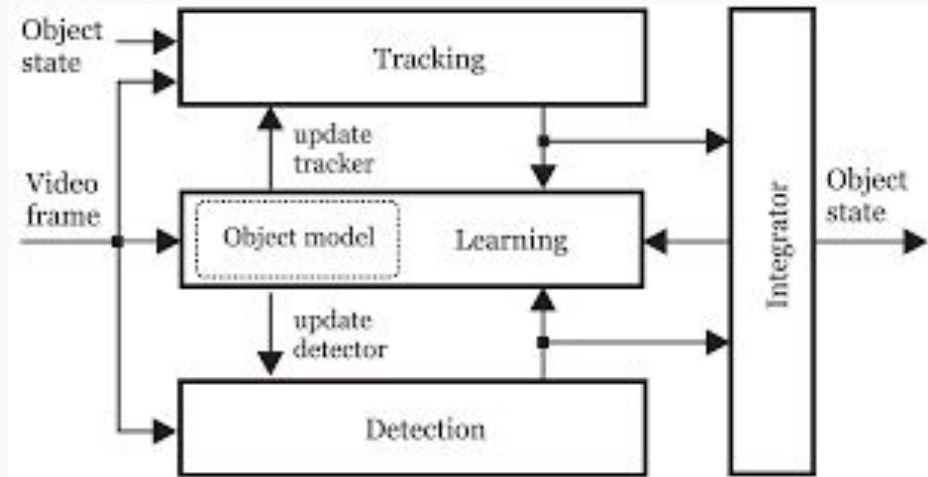
Other algo: MIL, TLD

Multiple Instance Learning



Tracking-Learning-Detection

Handle **long-term tracking** while simultaneously addressing issues like **drift** (when the tracker gets off-course) and **occlusion** (when the object is temporarily hidden).





Optical Flow

pattern of motion of objects in a video, captured as the movement of pixels between two consecutive frames.

cv2.calcOpticalFlowFarneback() can be used to compute dense optical flow

Histogram

distribution of pixel values (intensity or color) in an image.

used for tasks like brightness adjustment or contrast enhancement.

cv2.calcHist() to calculate histograms for frames.

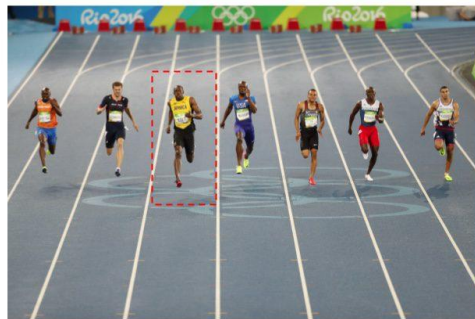




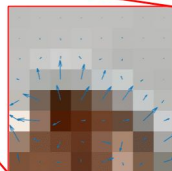
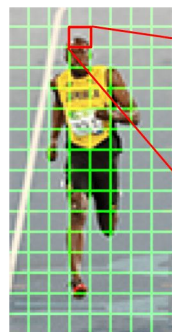
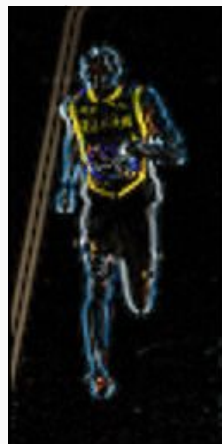
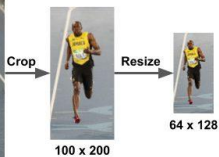
Histogram of Oriented Gradients

1. Gradient information: **Gradients** measure the change in intensity (brightness) of pixels in an image. Large changes often correspond to edges, which are important features for object recognition.
2. $G_x = I(x+1, y) - I(x-1, y)$ and $G_y = I(x, y+1) - I(x, y-1)$ I = intensity of the pixel
3. Edge orientation:

$$g = \sqrt{g_x^2 + g_y^2}$$
$$\theta = \arctan \frac{g_y}{g_x}$$



Original Image : 720 x 475



2	3	4	4	3	4	2	2
5	11	17	13	7	9	3	4
11	21	23	27	22	17	4	6
23	99	165	135	85	32	26	2
91	155	133	136	144	152	57	28
98	196	76	38	26	60	170	51
165	60	60	27	77	85	43	136
71	13	34	23	108	27	48	110

Gradient Magnitude

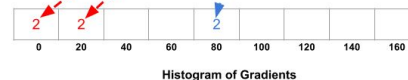
80	36	5	10	0	64	90	73
37	9	9	179	78	27	169	166
87	136	173	39	102	163	152	176
76	13	1	168	159	22	125	143
120	70	14	150	145	144	145	143
58	86	119	98	100	101	133	113
30	65	157	75	78	165	145	124
11	170	91	4	110	17	133	110

Gradient Direction

80	36	5	10	0	64	90	73
37	9	9	179	78	27	169	166
87	136	173	39	102	163	152	176
76	13	1	168	159	22	125	143
120	70	14	150	145	144	145	143
58	86	119	98	100	101	133	113
30	65	157	75	78	165	145	124
11	170	91	4	110	17	133	110

Gradient Direction

Gradient Magnitude



X, y and magnitude of gradient



Thresholding

technique used to segment an image by setting a threshold value for pixel intensity. Values above the threshold become one value (e.g., white), and values below become another (e.g., black)

`cv2.threshold()`

Frame Differencing

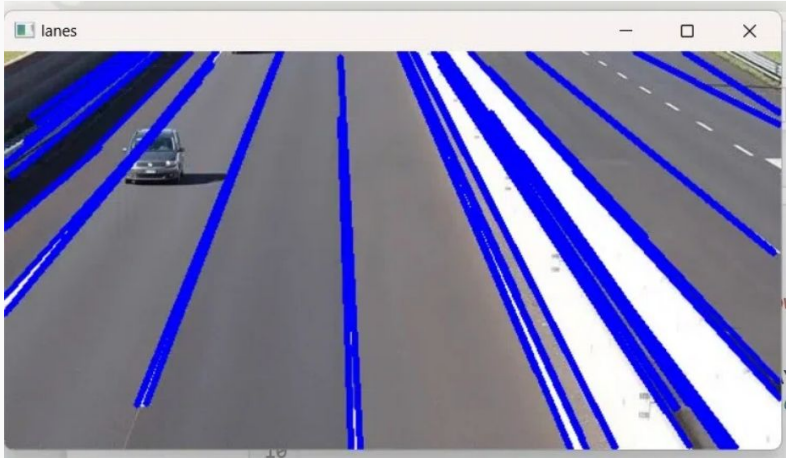
used for motion detection
comparing the difference between two consecutive frames
can detect moving objects

Hough Transform

used to detect shapes like lines, circles, and other objects in images and videos

edge detection and object recognition

cv2.HoughLines() or **cv2.HoughCircles()**

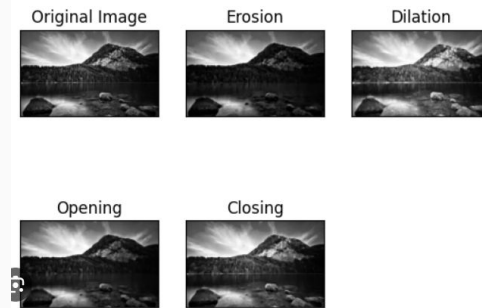
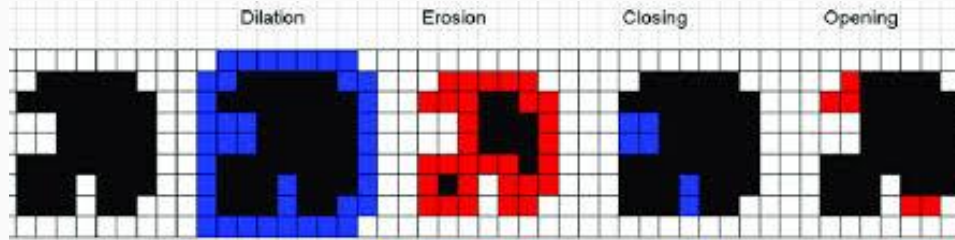


Morphological Operations

set of image processing techniques that process shapes within an image, often used for tasks like noise removal or object enhancement

erosion, dilation, opening, and closing.

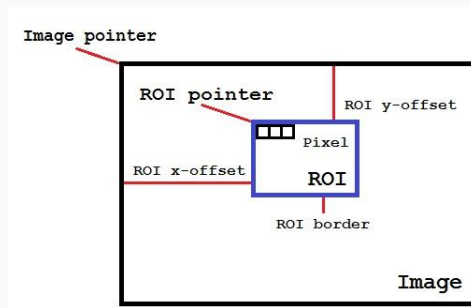
cv2.erode() and **cv2.dilate()**



ROI (Region of Interest)

specific region in a video frame that is selected for processing (e.g., face detection, object tracking).

slicing NumPy arrays representing video frames





Keyframes

selected frames in a video that represent significant changes or moments

keyframes are used as references for encoding the video.

identify keyframes by analyzing frame differences

Frame Buffer

memory region where the video frames are temporarily stored before processing or display.

Video Stream

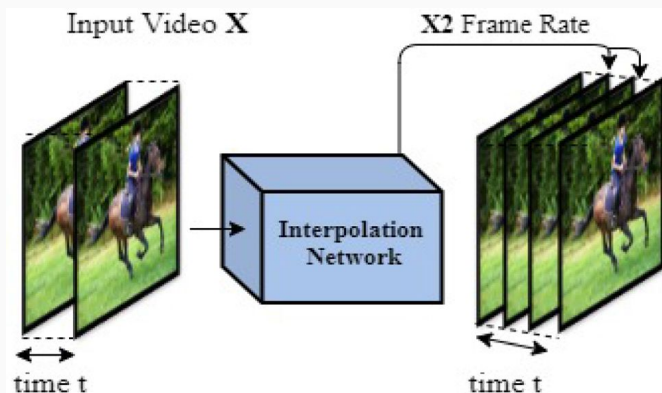
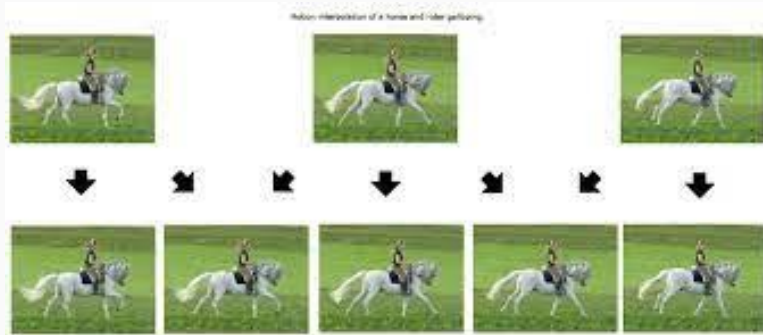
continuous flow of video data - any source

ideo streams using **cv2.VideoCapture()** to read the stream frame by frame.

Interpolation

resizing or transforming video frames

estimates new pixel values based on surrounding pixels



More reading:
<https://medium.com/@prabhs/creating-a-frame-interpolation-mode-l-from-scratch-using-convolutional-fusion-upsampling-5d0ed70a4447>

Nearest neighbour interpolation

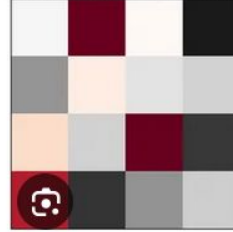
Diagram illustrating nearest neighbour interpolation. A blue arrow points from the first 3x3 grid to the second 6x6 grid, showing how the values are replicated.

10	4	22
2	18	7
9	14	25

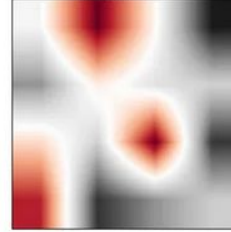
angeljohnsy.blogspot.com

10	10	4	4	22	22
10	10	4	4	22	22
2	2	18	18	7	7
2	2	18	18	7	7
9	9	14	14	25	25
9	9	14	14	25	25

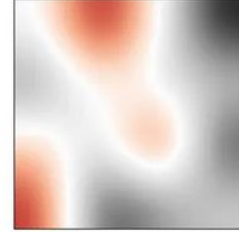
nearest neighbour



bilinear



bicubic

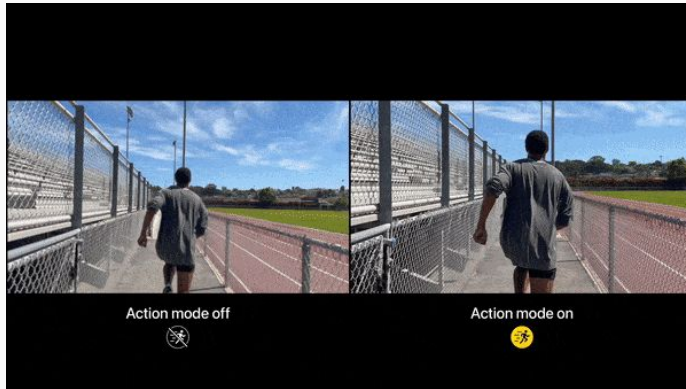




Stabilization

process of reducing or eliminating unwanted camera motion (e.g., shakes or jitters)

Stabilize video by analyzing the motion between consecutive frames and compensating for the motion using transformations





Spatial feature: local characteristics of an image or video frame that represent the arrangement of **pixels** in a particular region.

specific location or **local window** of the image

Edge, corner, textures and keypoints

Object detection and **recognition**, **Feature matching** across frames or **image stitching**.

Local Features:

Local features capture distinctive information around **specific points** or **regions** in the image **localized** within a **small region** of the image but are typically **invariant to scaling, rotation, and small changes in viewpoint**

Example: Tracking the **corner of a car** across multiple frames in a video to estimate motion.



Feature Type	Scope	Key Characteristics	Applications
Spatial Features	Local region of interest	Describes pixel arrangements and edges in a specific region.	Object detection, edge detection, keypoint extraction.
Local Features	Specific points/regions	Invariant to scale/rotation, robust features for matching/movement.	Object tracking, matching, image stitching.
Global Features	Entire image/scene	Describes overall characteristics of the image (e.g., texture, color).	Image classification, scene recognition.
Temporal Features	Across time/frames	Captures the dynamics, motion, and evolution of objects across time.	Action recognition, motion tracking.

Stabilizing Algorithms	Key Characteristics	Best For
Global Motion Estimation (GME)	Estimates and compensates for global motion in video.	General-purpose stabilization.
Warping / Spatial Filtering	Applies geometric transformations to minimize motion.	Handling large motion variations and aligning scenes.
Local Motion Estimation	Tracks motion on a local scale rather than global.	Videos with localized motion or when fine-grained stabilization is needed.
Frame Blending / Interpolation	Smooths transitions by blending or interpolating between frames.	Removing abrupt shakes or creating smooth transitions.
Temporal Filtering	Applies smoothing techniques over a series of frames.	Videos with minor shaking or jitter.
Feature-Based Tracking	Tracks specific features and compensates for motion.	Object-based stabilization and tracking.
Deep Learning-Based Stabilization	Uses CNNs to predict motion and apply corrections.	Complex videos with varied or unpredictable motion.



Video Streaming

transmit video data

OpenCV, ffmpeg, or GStreamer

external library - moviepy

FFmpeg

video processing, encoding, decoding, and streaming

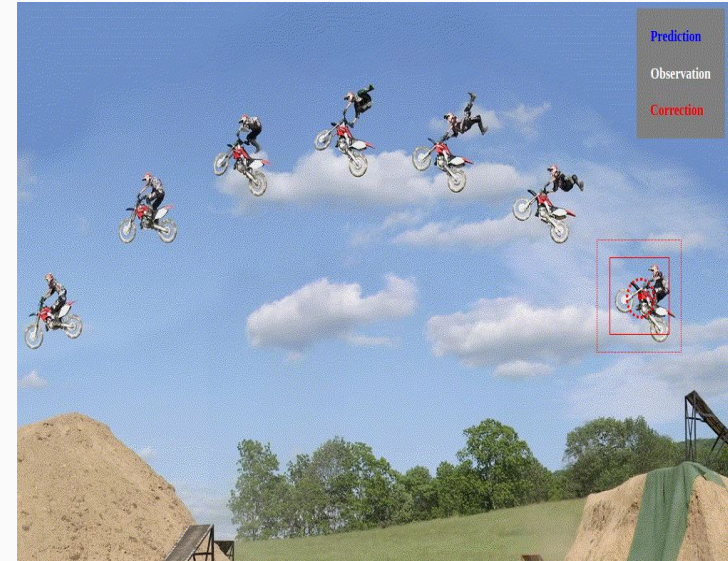
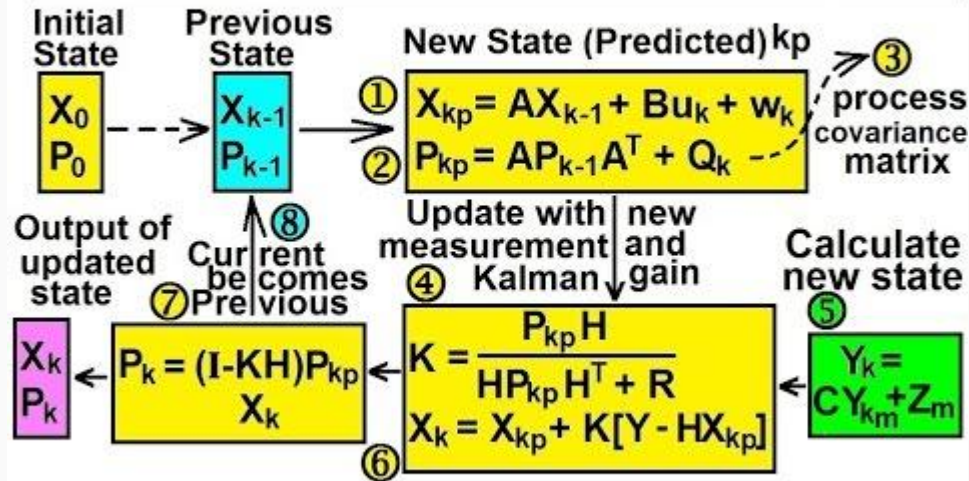
Ffmpeg-pyth

GOP (Group of Pictures)

collection of frames used in video compression

typically includes I-frames (keyframes) and P-frames (predicted frames)

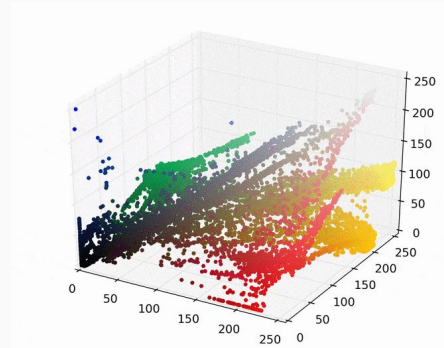
The Kalman Filter: Flow Chart 2D: Tracking Plane



Kalman Filter

optimal estimation technique used for predicting the state of a system from noisy, uncertain, or incomplete observations. It combines information from a system's dynamics model (how the system evolves over time) and the measurement model (how observations relate to the true state) to estimate the true state with minimal error

Meanshift - non-parametric clustering



Camshift:



Mean shift window
initialization